UNIT 1: UNITS AND MEASUREMENTS

A quick, color-coded rewrite of the handwritten notes

Physical quantities & measurement

Physics is a quantitative science: we measure physical properties in experiments and everyday life. A measurement compares a quantity with an accepted standard unit.

Measurement = numerical value × unit
Example: 5 m means a length with magnitude 5, measured in metres.

Units

A unit is the standard used to measure a physical quantity. Different quantities use different units.

Fundamental quantities

Fundamental quantities do not depend on other physical quantities for their measurement. The seven SI base quantities are:

Quantity	SI unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Electric current	ampere	A
Luminous intensity	candela	cd
Amount of substance	mole	mol

Derived quantities & units

A derived quantity depends on one or more other physical quantities. A derived unit is formed from base units.

Quantity	SI unit	Symbol
Area	square metre	m^2
Volume	cubic metre	m^3
Density	kilogram per cubic metre	kg/m^3
Velocity	metre per second	m/s
Force	newton	$N = kg \cdot m/s^2$
Energy / work	joule	J
Power	watt	W

Supplementary quantities in the notes

Quantity	SI unit	Symbol
Plane angle	radian	rad
Solid angle	steradian	sr

Common systems of units

- CGS: centimetre, gram, second
- FPS: foot, pound, second
- MKS: metre, kilogram, second
- SI: International System of Units; the standard system used for science and measurement

Scalar vs vector

Type	What describes it?	Examples
Scalar	Magnitude only	length, mass, time, temperature, speed, energy
Vector	Magnitude and direction	displacement, velocity, acceleration, force, weight

Dimensions & dimensional formula

The dimensions of a quantity show the powers of base quantities needed to express it. A dimensional formula writes a derived quantity using base symbols in square brackets.

Base dimensions: length [L] • mass [M] • time [T] • temperature [Θ] • current [I] • luminous intensity [J] • amount [N]

The SI base units are m, kg, s, K, A, cd and mol. Dimensional symbols: L, M, T, Θ, I, J and N.

Quantity	SI unit	Dimensional formula
Area	m^2	$[L^2]$
Volume	m^3	$[L^3]$
Density	kg/m^3	$[M\ L^{-3}]$
Velocity	m/s	$[L\ T^{-1}]$
Acceleration	m/s^2	$[L\ T^{-2}]$
Force	N	$[M\ L\ T^{-2}]$
Momentum	$kg·m/s$	$[M\ L\ T^{-1}]$
Energy / work	J	$[M\ L^2\ T^{-2}]$
Power	W	$[M\ L^2\ T^{-3}]$
Pressure	N/m^2	$[M\ L^{-1}\ T^{-2}]$

Errors in measurement

Measurement results have uncertainty. The notes group errors into two broad types:

- Systematic errors: caused by a consistent problem in the instrument, method, setup or observer. Examples include a faulty meter, a consistent reading mistake, parallax, or a graduated tube held at an angle.
- Random errors: unpredictable variations that remain after careful procedure. They cannot be removed completely; repeated readings and averaging help reduce their effect.

Reduce systematic error: use a suitable, correctly working instrument; follow the correct procedure; read scales carefully and avoid parallax.

Estimating error from repeated readings

For observations $a_1, a_2, \ldots, a_n$:

Mean: $a_{mean} = (a_1 + a_2 + \ldots + a_n) / n$

Absolute error for each reading: $|\Delta a_i| = |a_{mean} - a_i|$

Mean absolute error: $\Delta a_{mean} = (|\Delta a_1| + |\Delta a_2| + \ldots + |\Delta a_n|) / n$

Relative error = $\Delta a_{mean} / a_{mean}$

Percentage error = $(\Delta a_{mean} / a_{mean}) \times 100\%$

Worked error example

Readings: 1.1, 1.2 and 1.3

Step	Calculation	Result
Mean	(1.1 + 1.2 + 1.3) / 3	1.2
Absolute errors	$\lvert 1.2-1.1 \rvert$, $\lvert 1.2-1.2 \rvert$, $\lvert 1.2-1.3 \rvert$	0.1, 0, 0.1
Mean absolute error	(0.1 + 0 + 0.1) / 3	0.067
Relative error	0.067 / 1.2	≈ 0.056
Percentage error	0.056 × 100%	≈ 5.6%

A measurement written as 50 ± 2 cm has percentage error = (2 / 50) × 100% = 4%.

Significant figures

Significant figures are the meaningful digits in a measured number.

- All non-zero digits are significant. Example: 347 has 3 significant figures.
- Zeros between non-zero digits are significant. Example: 2004 has 4.
- Trailing zeros after a decimal point are significant. Example: 2.1000 has 5.
- Trailing zeros in a whole number without a decimal point are not assumed significant. Example: 100000 may show only 1 significant figure.
- Leading zeros in a number below 1 are not significant. Example: 0.0012 has 2.
- In scientific notation, count the digits in the coefficient; the power of ten does not add significant figures. Examples: $1 \times 10^{\square}$ has 1; 6.2×10^{3} has 2.

Measurements in daily life

- Medical checks: temperature, pulse and blood counts.
- Home appliances: temperature and time settings help cook or heat food.
- Transport: fuel, air pressure, oil and speed are measured.
- Shopping: food, groceries, vegetables and clothing are measured.

Measurements are part of everyday decisions, science and technology.

Ancient astronomical instruments

The notes describe instruments used in India to estimate large distances and heights, measure angles, track star positions and tell time. Bhaskaracharya (Bhaskara II), an Indian mathematician and astronomer, designed instruments and wrote Siddhanta Shiromani.

Instrument	What the notes say it was used for
Chakra yantra (wheel)	A disc-like protractor for marking angles and locating celestial positions; also used to estimate time from shadows and altitude.
Dhanur yantra (bow)	A semicircular instrument for vertical angles and estimating heights, Earth / Moon dimensions and distances.
Yashti yantra (staff)	A cross-staff used to calculate the height of large objects such as trees or hills.
Phalaka yantra (board)	A board used to find time from the Sun's altitude.

End of Unit 1 notes

UNIT 2: ELECTRICITY, MAGNETISM & SEMICONDUCTORS

Charge, electric fields, potential and field lines

Electric charge

Charge is a basic property of particles. Protons are positive, electrons negative, and neutrons neutral. The SI unit is the coulomb (C).

Elementary charge: $e = 1.6 \times 10^{-19}$ C • Quantization: $Q = n\,e$

- Additivity: total charge is the algebraic sum of the charges; include their signs.
- Conservation: charge can transfer between bodies, but the total charge of an isolated system stays constant.
- Interaction: like charges repel; unlike charges attract.

Coulomb's inverse-square law

For two stationary point charges q_1 and q_2 separated by distance r, force magnitude rises with the product of the charges and falls with r^2.

$$F = (1 / 4\pi\varepsilon) \times |q_1 q_2| / r^2$$
$$\text{In vacuum: } F = (1 / 4\pi\varepsilon_0) \times |q_1 q_2| / r^2 \approx 9 \times 10^9 |q_1 q_2| / r^2$$

ε is the permittivity of the medium; ε_0 is the permittivity of free space.

Electric field & intensity

An electric field is the region where a charge experiences electric force. Field intensity is force per unit positive test charge: $E = F/q$. For a point charge Q: $E = (1 / 4\pi\varepsilon_0) Q/r^2$.

E is a vector. SI units: N/C or V/m. A uniform field has the same size and direction everywhere (for example, between parallel plates); a point charge produces a non-uniform field.

Potential and potential difference

Electric potential is work per unit positive charge from a reference point: $V = W/q$. For a point charge, $V = (1 / 4\pi\varepsilon_0) Q/r$. Potential difference is work per unit charge moved between two points.

1 volt = 1 joule / 1 coulomb • Conventional current flows from higher to lower potential.

ELECTRIC FIELD LINES & MAGNETISM

Use field lines to picture field direction and strength

Electric lines of force

An electric field line is an imaginary curve whose tangent gives the field direction at each point.

- Lines start on positive charges and end on negative charges.
- They never cross or intersect.
- They meet a conductor surface at right angles; in electrostatic equilibrium, the field inside a conductor is zero.
- They can pass through insulators.
- Crowded lines mean a stronger field.
- Uniform-field lines are parallel and equally spaced.

Magnetism

Magnetism is the attraction or repulsion associated with magnets and magnetic materials. A freely suspended bar magnet points roughly north–south. Like poles repel; unlike poles attract. Isolated magnetic poles (monopoles) are not observed.

- Natural magnetite is a natural magnet; magnets can also be made artificially. Uses include motors, bells, speakers, fans, lifting iron and refrigeration.

Magnetic field & intensity

A magnetic field is the region where magnetic force acts. Magnetic field intensity H describes field strength; the notes give SI unit A/m. H is a vector.

Magnetic field lines

- Lines form continuous closed loops and do not intersect.
- Outside a magnet they go from north to south; inside, south to north.
- Closer lines indicate a stronger field. The tangent gives direction at a point.

Magnetic flux: $\Phi = B\,A\cos\theta$ • SI unit: weber (Wb)
Flux density: $B = \Phi/A$ (for a perpendicular field) • SI unit: tesla (T)

$B = \mu H$, where μ is permeability. 1 tesla = 10□ gauss.

CURRENT, OHM'S LAW & MATERIAL PROPERTIES

Core circuit quantities and their units

Electric current

Current is the rate of flow of charge: $I = Q/t$. SI unit: ampere (A). One ampere means one coulomb passes a point each second: 1 A = 1 C/s. Conventional current follows positive-charge flow, from + to −.

Ohm's law

At constant physical conditions, current through a conductor is directly proportional to the potential difference across it.

$$V = I R \bullet R = V/I \bullet 1\ \Omega = 1\ V/A$$

Resistance opposes current; SI unit is ohm (Ω). For an ohmic conductor, the I–V graph is a straight line through the origin.

Conductance, resistivity & conductivity

Quantity	Meaning / formula	SI unit
Conductance G	Reciprocal of resistance: $G = 1/R$	siemens (S)
Resistivity ρ	$\rho = R A / \ell$	ohm metre (Ω·m)
Conductivity σ	Reciprocal of resistivity: $\sigma = 1/\rho$	S/m

A longer conductor has more resistance; a larger cross-sectional area reduces resistance: $R = \rho\ell/A$.

Joule heating

Current through resistance converts electrical energy into heat. The heating effect is used in irons, heaters and geysers.

$$\text{Joule's law: } H = I^2 R\, t \bullet \text{Also } H = V\, I\, t = (V^2/R)t$$

RESISTOR NETWORKS & GALVANOMETERS

Series, parallel and measurement devices

Resistors in series

A single path carries the same current through every resistor. Supply voltage divides among them.

$$R_{eq} = R_1 + R_2 + \dots + R_n$$

Resistors in parallel

Each branch connects across the same two points, so voltage is the same across each branch and total current splits among branches.

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

Galvanometer

A galvanometer detects small currents. In a moving-coil galvanometer, a current-carrying coil in a magnetic field experiences torque and deflects.

Convert to	How to connect	Added resistance
Ammeter	Low shunt in parallel	$R_{sh} = I_g R_g / (I - I_g)$
Voltmeter	High resistance in series	$R = V/I_g - R_g$

I_g = galvanometer full-scale current; R_g = galvanometer resistance; I = desired ammeter range; V = desired voltmeter range.

Ammeter: very low resistance, connected in series. Voltmeter: very high resistance, connected in parallel across the component.

CONDUCTORS & ENERGY BANDS

Why solids conduct differently

In a solid, valence-band and conduction-band energy levels form bands. The gap between them is the forbidden energy gap Eg; electrons need enough energy to cross it.

Material	Band picture	Typical behaviour
Conductor	Bands overlap or no significant gap	Many mobile electrons; low resistance
Semiconductor	Small forbidden gap	Conductivity between conductor and insulator; rises with temperature
Insulator	Large forbidden gap	Very few mobile carriers; high resistance

Conductors

- Allow charge to flow easily; examples: copper, aluminium, gold and silver.
- High conductivity and low resistivity. Resistance generally rises with temperature for metals.

Semiconductors

- Conductivity lies between conductors and insulators. Examples: silicon (Si) and germanium (Ge).
- A small energy gap lets heat or light free some electrons; conductivity generally increases with temperature.

Insulators

- Strongly resist charge flow; examples: wood, rubber and plastic.
- They have very low conductivity and a large gap between valence and conduction bands.

Band names: valence band = normally bound electrons; conduction band = mobile electrons; forbidden gap = energy range with no allowed states.

INTRINSIC & EXTRINSIC SEMICONDUCTORS

Pure crystals and doping

Intrinsic (pure) semiconductor

An undoped semiconductor such as pure silicon or germanium. Each atom forms covalent bonds with neighbours. At very low temperature it behaves nearly as an insulator; as temperature rises, more electron–hole pairs form.

Extrinsic (doped) semiconductor

Adding a small, controlled impurity to a pure semiconductor increases conductivity. The host is Si or Ge; the added impurity is the dopant.

Type	Dopant	Majority carrier	Examples
P-type	Trivalent acceptor	Holes	B, Al, In
N-type	Pentavalent donor	Electrons	P, As, Sb

P-type

A trivalent impurity has three valence electrons, leaving a bond short of one electron. This vacancy behaves as a hole. Holes are majority carriers; electrons are minority carriers.

N-type

A pentavalent impurity has five valence electrons. Four take part in bonds; the fifth is relatively free. Electrons are majority carriers; holes are minority carriers.

Acceptor doping → P-type → holes dominate.
Donor doping → N-type → electrons dominate.

P–N JUNCTION DIODES

Biasing, I–V behaviour and uses

Junction and depletion region

Joining P-type and N-type material creates a P–N junction. Electrons and holes diffuse across the boundary and recombine, leaving a region depleted of mobile carriers. This creates a potential barrier.

Forward and reverse bias

Bias	Connection	Junction behaviour
Forward	P to +, N to −	Depletion region narrows; current rises after the cut-in voltage
Reverse	P to −, N to +	Depletion region widens; tiny leakage current until breakdown

Approximate cut-in voltage: silicon ≈ 0.7 V; germanium ≈ 0.3 V.

Diode resistance & reverse breakdown

Static resistance: Rdc = Vf/If. Dynamic resistance: Rac = ΔVf/ΔIf. In reverse bias, a large enough voltage can cause breakdown and a sharp current rise.

Applications

- Switching in logic circuits; clipping and clamping circuits; voltage multipliers and regulators.
- Rectification: converts AC to DC. A half-wave rectifier passes one half-cycle and blocks the other.

UNIT 3: THERMOMETRY & FIBER OPTICS

Heat, temperature and temperature scales

Heat & temperature

Heat is energy transferred from a hotter body to a colder one. Temperature measures how hot or cold a body is and is related to the average kinetic energy of its particles.

Quantity	SI unit	Other units / note
Heat	joule (J)	1 cal ≈ 4.184 J; 1 kcal = 1000 cal
Temperature	kelvin (K)	degree Celsius (°C), degree Fahrenheit (°F)

- Heat is an extensive quantity: it depends on the amount of material. Temperature is intensive: it does not depend on sample size.

Heat is the cause; temperature change is an effect.

Temperature conversions

K = °C + 273.15 (often rounded to 273 in class problems)
°F = (9/5)°C + 32 • °C = (°F − 32) × 5/9

Example	Conversion	Answer
100°C	K = 100 + 273; °F = 1.8×100 + 32	≈ 373 K; 212°F
98.6°F	°C = (98.6−32)×5/9; K ≈ °C + 273	37°C; ≈ 310 K
300 K	°C ≈ 300−273; °F = 1.8×27 + 32	27°C; 80.6°F

The Kelvin scale begins at absolute zero (0 K ≈ −273.15°C). Water freezes at 0°C / 32°F and boils at 100°C / 212°F at standard atmospheric pressure.

HEAT TRANSFER

Conduction, convection and radiation

Heat flows spontaneously from a higher-temperature region to a lower-temperature region. There are three main modes:

Mode	How heat moves	Where it occurs / example
Conduction	Energy passes between particles without bulk movement	Mainly solids; a metal spoon warms in hot tea
Convection	Heat moves with the actual motion of fluid particles	Liquids and gases; warm air rises
Radiation	Energy travels as electromagnetic waves; no material medium needed	Sunlight warming Earth

Conductors and insulators of heat

- Good conductors: heat passes through readily; examples include copper, iron, silver, aluminium and steel.
- Insulators (bad conductors): slow heat transfer; examples include wood, plastic and glass.

Thermal conductivity

For steady conduction through a uniform rod, heat transferred Q increases with area A, temperature difference ($\theta_1 - \theta_2$), and time t; it decreases as length d increases.

$$Q = k\,A\,(\theta_1 - \theta_2)\,t / d$$

k is the coefficient of thermal conductivity. SI unit: W/(m·K), equivalent to J/(m·K·s).

GAS LAWS & THE IDEAL GAS EQUATION

For a fixed amount of gas

Boyle's law — constant temperature

Pressure varies inversely with volume: when volume decreases, pressure increases.

$P \propto 1/V$ • $PV = \text{constant}$ • $P_1V_1 = P_2V_2$

Charles's law — constant pressure

Volume is directly proportional to absolute temperature.

$V \propto T$ • $V/T = \text{constant}$ • $V_1/T_1 = V_2/T_2$

Gay-Lussac's law — constant volume

Pressure is directly proportional to absolute temperature.

$P \propto T$ • $P/T = \text{constant}$ • $P_1/T_1 = P_2/T_2$

Use kelvin for gas-law temperatures: $T(K) = t(°C) + 273.15$.

Ideal gas equation

$$PV = nRT$$

P = pressure, V = volume, n = amount in moles, T = absolute temperature, and R = universal gas constant ($8.314\ J \cdot mol^{-1} \cdot K^{-1}$).

- These relationships describe how pressure, volume and temperature change together for a fixed amount of gas.

SPECIFIC HEAT CAPACITY

How much heat raises a substance's temperature?

Specific heat capacity, c

Specific heat capacity is the heat required to raise the temperature of 1 kg of a substance by 1 K (or 1°C). Different materials warm by different amounts for the same heat input.

$$Q = m\,c\,\Delta T \bullet c = Q / (m\,\Delta T)$$

Q = heat transferred; m = mass; ΔT = temperature change. SI unit of c: J/(kg·K).

Specific heats of gases

Symbol	Condition	Meaning
Cp	Constant pressure	Heat capacity per unit mass (or per mole) at fixed pressure
Cv	Constant volume	Heat capacity per unit mass (or per mole) at fixed volume

- For a gas, Cp is greater than Cv: at constant pressure, some supplied energy also does expansion work.

For molar heat capacities: $C_p - C_v = R$.
Ratio of specific heats: $\gamma = C_p/C_v$.

The source notes use c for specific heat and Q for heat. Keep mass and units consistent when substituting into $Q = mc\Delta T$.

REFLECTION, REFRACTION & TOTAL INTERNAL REFLECTION

Light changes direction at surfaces and boundaries

Reflection

Reflection occurs when light strikes a surface and returns in the same medium.

> **Law of reflection: angle of incidence i = angle of reflection r.**
> **Incident ray, reflected ray and the normal lie in the same plane.**

Refraction

Refraction is the change in direction of light as it enters a different transparent medium, caused by a change in speed.

- From a rarer to a denser medium, a ray bends toward the normal; from denser to rarer, it bends away.

> **Snell's law: $n_1 \sin i = n_2 \sin r$**
> **Refractive index (absolute): $n = c/v$**

Total internal reflection (TIR)

When light travels from a denser medium to a rarer medium, the angle of refraction grows with the angle of incidence. At the critical angle ic, the refracted ray is at 90°; for i > ic, all light reflects back into the denser medium.

> **TIR conditions: (1) denser to rarer medium; (2) incidence angle greater than the critical angle.**
> **For $n_1 > n_2$: $\sin ic = n_2/n_1$.**

OPTICAL FIBERS

Structure, working principle and applications

What is an optical fiber?

A very thin transparent fiber carries light signals over distance with little loss. It works by repeated total internal reflection.

Part	Role
Core	Inner glass or plastic path; higher refractive index
Cladding	Surrounds the core; lower refractive index keeps light inside
Jacket	Outer protective covering

How it guides light

Light enters the core within an allowed range. At the core–cladding boundary it meets the TIR condition, reflects repeatedly, and travels along the fiber.

Acceptance angle θa: largest entry angle (measured from the fiber axis) that still lets light propagate.
For air outside: NA = sin θa = √(ncore² − ncladding²).

Fiber types

Classification	Types	Typical use
Refractive index	Step-index; graded-index	Graded index changes gradually from core axis outward
Light propagation	Single-mode; multimode	Single-mode: narrow core, laser, long distance. Multimode: wider core, LED, shorter distance.
Material	Glass; plastic	Chosen for the application and required transmission

Applications

- High-speed internet and computer networks; cable television.
- Medicine and research, including endoscopy; mechanical inspection.
- Automotive lighting, decorative lighting and communications.